MASTER IN ASTROPHYSICS AND SPACE SCIENCE

TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Optimisation & Monte Carlo Sampling

Introduction to Numerical Methods for Astrophysics

Fritz Ali Agildere

November 27, 2025

Overview

Probability Refresher

- Definitions & Notation
- Random Numbers

Optimisation

- Common Methods
- Problems

Markov Chain Monte Carlo

- Popular Algorithms
- Tuning & Diagnosing
- Alternatives

Overview

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Frequentist View

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- *Without Prior Knowledge:* out of n independent measurements the outcome A occurs k times, defining

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Bayes' Theorem

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

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- *Variance* or second raw moment: $\sigma^2 = E\left[(X - E[X])^2\right] = \int_{-\infty}^{+\infty} \left(x - E[x]\right)^2 f(x)\, dx$

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- *Arbitrary Distributions:* obtained from uniformly distributed numbers by various methods

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$$I_\nu(T,\beta) = B_\nu(T)\left(1 - e^{-\tau_\nu(\beta)}\right) \approx B_\nu(T)\tau_\nu(\beta) = B_\nu(T)\Sigma\kappa_\nu(\beta) = \frac{2h\Sigma\kappa_0}{c^2{\nu_0}^\beta}\frac{\nu^{3+\beta}}{\exp\left(\frac{h\nu}{kT}\right) - 1}$$

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  - suppose we have no prior information on the parameter distribution

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- *Likelihood:*

$$L(I_\nu|v,\sigma_\nu,\theta) = \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi\sigma_{\nu,i}^2}} \exp\left(-\frac{\left(I_{\nu,i} - I_\nu(\theta)\right)^2}{2\sigma_{\nu,i}^2}\right)$$

Example

- *Likelihood:*

$$\log L(I_\nu | \nu, \sigma_\nu, \theta) = -\frac{N}{2}\log(2\pi) - \sum_{i=1}^{N} \sigma_{\nu,i} - \frac{1}{2}\sum_{i=1}^{N} \frac{\left(I_{\nu,i} - I_\nu(\theta)\right)^2}{\sigma_{\nu,i}^2}$$

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- *Objective:* minimise the $\chi^2(\theta)$ measure to maximise the posterior probability

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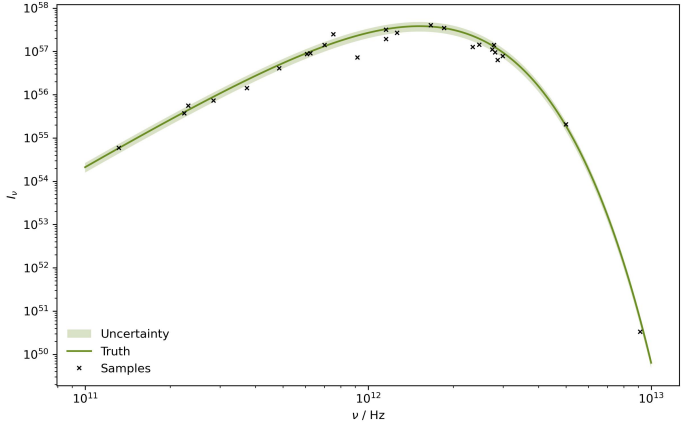
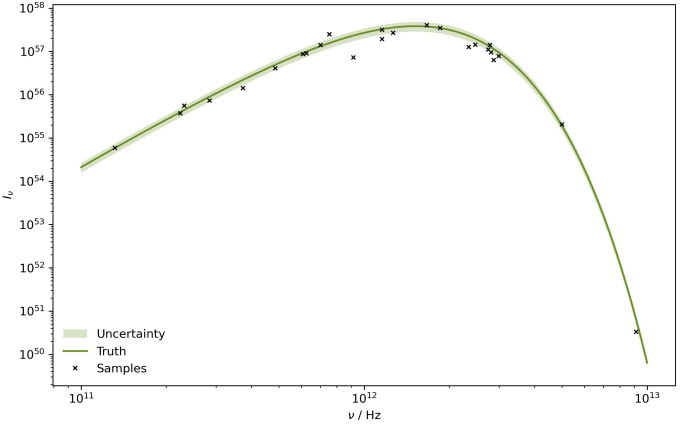


Measurements

How do we proceed from this dataset to maximise the likelihood for our nonlinear model?



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- *Problem:* as D increases we have N^D gridpoints, and with more complex parameter distributions, this procedure quickly becomes intractable due to computational limits

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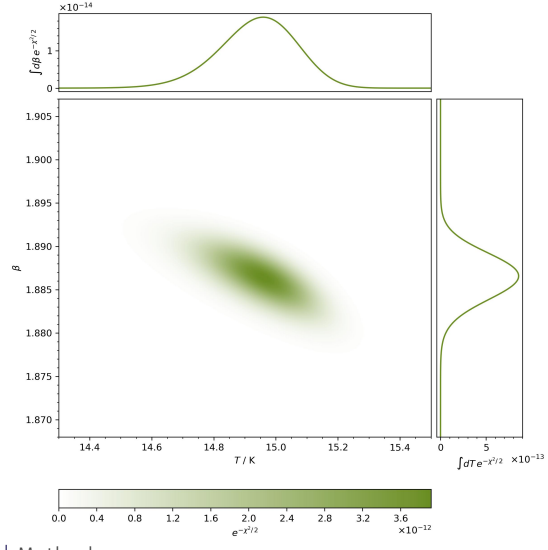
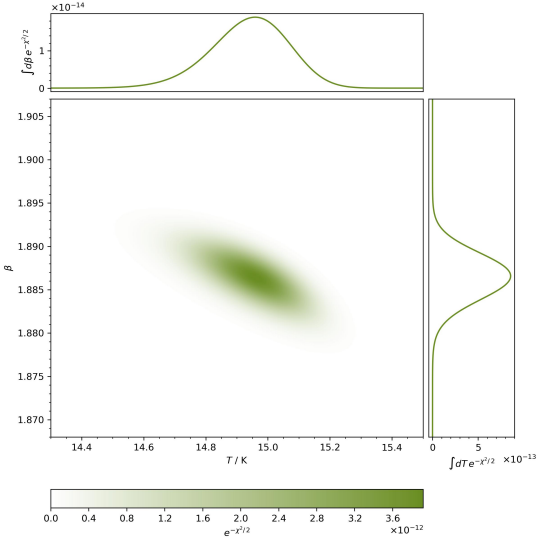


Parameter Grid

Can we think of more elegant ways to explore even very high dimensional parameter spaces?



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- *Interactive Visualisations:* check out Ben Frederickson[1] and Andy Casey[2] to see some nice examples

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  - improve by considering last step and optimizing step range

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  - typical settings are $\tau = \kappa = 0.5$ (L. Armijo, *Pac. J. Math.* 1966)

Thank You!